

THE CRISIS

WATER

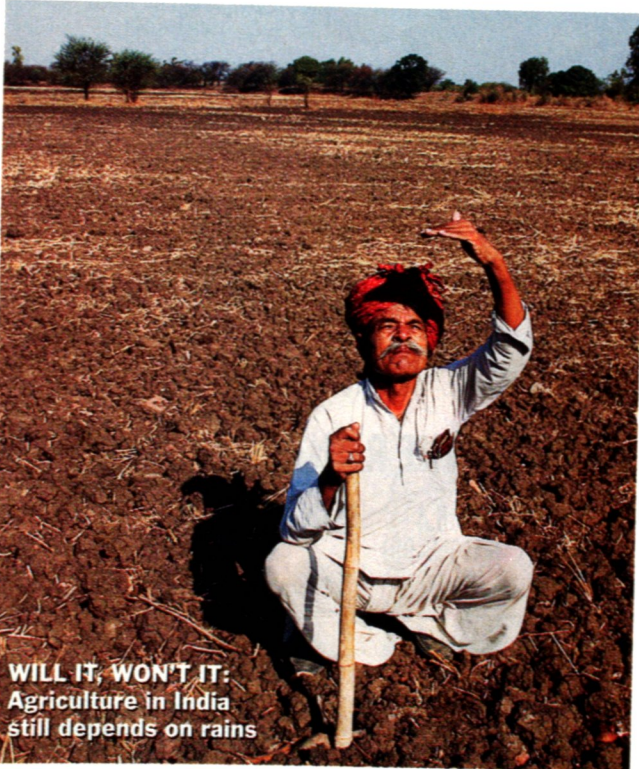
- India commands just 4 per cent of the global freshwater resources, but supports 16 per cent of world population
- Adverse effects of droughts on production of crops
- Over-dependence on rainfall and monsoon. More than 60 per cent of the cultivated area is rain-dependent. The country's rainfall is not evenly distributed, but in total, it is adequate to meet the water requirement
- Over-exploitation of water, especially for growing rice and sugar, has seen water tables recede

THE CONTROL

- Creating irrigation potential, repairing system deficiencies and inefficient on-farm water management. Adapting cropping patterns according to water availability
- Reducing the utilisation gap, ground water extraction, watershed development, rainwater harvesting
- Increasing irrigation efficiency from the present level of 35-40 per cent, completion of ongoing irrigation projects. River grid development will help rationalise the availability of water
- National Rainfield Authority to help conserve water

50% fall in the rate of additional land brought under irrigation.

PANKAJ TIWARI



WILL IT, WON'T IT:
Agriculture in India
still depends on rains

the potential added was only half of the stated figure of 16 million hectare. Meanwhile, there are some 480-odd irrigation projects that need to be completed apart from 240 new ones being proposed. As Y.K. Alagh, a former Union minister of state and an expert on agriculture, points out, "We didn't seem to understand the crisis of land and water till a year ago even though public investment had fallen and agriculture growth rates had flattened."

Meanwhile, the over-exploitation of water, especially for growing rice and sugar, has seen water tables drop dramatically. A recent report of the Punjab Farmers Commission found that the water table had reached the critical depth of more than 10 meter in as much as 85 per cent areas in central Punjab, which has sweet ground water.

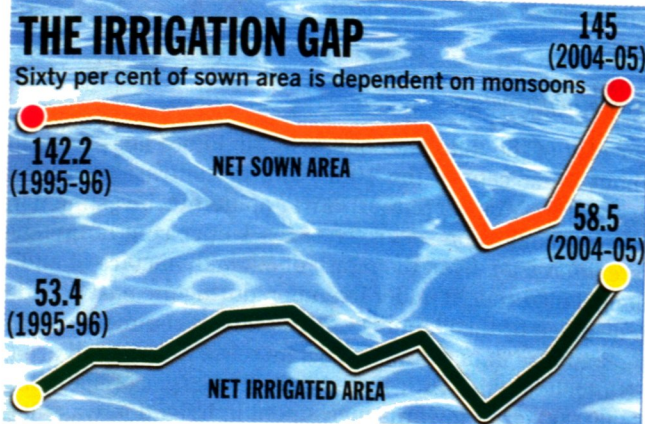
G.S. Kalkat, chairman of the Commission, says, "The national food security is important but it cannot be at the cost of imperilling Punjab's ground water resources beyond redemption." The Punjab Government has been focusing its diversification plan on shifting at least one million hectare under paddy to other less water-consuming crops. Dr S.S. Johl, a farmer-economist of repute, warns: "If the current trend of decline in the ground water level by one meter a year continues, Punjab could turn into a barren territory in next two decades."

Other states too face similar problems. In Karnataka, ground water levels were depleting at an alarming rate of 70 cm a year. In Uttar Pradesh, three years of drought have brought farmers to their knees. "Agle ek do saalon mein agar yahan bhi Bundelkhand aur Vidarbha jaisi atmahatyaen ho to aap ashcharya nahin kariyega (Don't be surprised if the farmers of this region too start committing suicide like the ones from Bundelkhand and Vidarbha)," lamented Prem Prakash Tewari, a farmer in Sarwan village in Unnao district who was given a paltry sum of Rs 65 as a drought relief last year.

The Centre has now set up a National Rainfield Authority to find out how best to conserve water in these regions and to come up with a time-bound plan. However, it would take years for the country to boost irrigation facilities and implement an effective watershed management programme.

THE IRRIGATION GAP

Sixty per cent of sown area is dependent on monsoons



Figures in million hectare